PHYSICS 317: a challenge problem on Lagrange's equations — solutions

Challenge problem (optional): a free particle in an expanding universe. In Newtonian cosmology, space is filled with matter of uniform density $\rho(t)$, and every distance grows by one common factor: a particle that just rides the expansion sits at $r = a(t)\,x$ with x fixed, where $a(t)$ is the *scale factor* and x is its *comoving coordinate*. By Newton's shell theorem the matter inside radius r pulls a particle there as if all of it sat at the origin (and the matter outside exerts no net force), so a particle of mass m at distance r from the origin has potential energy

$$U(r,t) = \frac{2\pi G}{3}\, m\rho(t)\, r^2 \qquad \Big[\text{check: } -\frac{\partial U}{\partial r} = -\frac{Gm}{r^2}\cdot\frac{4\pi}{3}\rho r^3 = -\frac{GmM(<r)}{r^2}\Big],$$

and the scale factor obeys the Friedmann equation $\ddot{a} = -\frac{4\pi G}{3}\rho\, a$ (Newton's second law for a particle riding the expansion, with $r = ax$; you may take it as given). Now consider a particle of mass m that moves along a line through the origin but does *not* necessarily ride the expansion.

(a) Write down the Lagrangian $\mathcal{L}(r,\dot{r},t)$. Find the Lagrange equation for r and check that it is Newton's second law with the force above. Explain why neither of our two conservation tests (an ignorable coordinate, Section 7.6; no explicit t in $\mathcal{L}$, Section 7.8) hands you a conserved quantity here.

(b) Change to the comoving coordinate $x = r/a(t)$ (a generalized coordinate whose relation to r involves t, as in Example 7.6). Show that

$$\mathcal{L} = \tfrac{1}{2}ma^2\dot{x}^2 + \frac{d}{dt}\Big(\tfrac{1}{2}ma\dot{a}\,x^2\Big).$$

Hint: expand the square in $\frac{1}{2}m\dot{r}^2$; write the cross term $ma\dot{a}\,x\dot{x}$ as a total time derivative minus what is left over; use the Friedmann equation on what is left over.

(c) Explain why the total time derivative can be dropped without changing the equation of motion (Problem 7.48, or argue directly from Hamilton's principle). From here on take $\mathcal{L} = \frac{1}{2}ma(t)^2\dot{x}^2$.

(d) Which coordinate is now ignorable? Find the conserved generalized momentum, and show that the *peculiar velocity* $v_{\text{pec}} \equiv a\dot{x}$, the particle's velocity relative to the expansion, falls as $1/a$. (When the universe doubles in size, a galaxy's random motion halves.)

(e) Find the Hamiltonian $\mathcal{H} = p\dot{x} - \mathcal{L}$. Is it $T+U$? Is it conserved? Use $d\mathcal{H}/dt = -\partial\mathcal{L}/\partial t$ (Section 7.8) to find how $\mathcal{H}$ depends on a.

(f) General relativity gives the same conservation law for light: the momentum a local observer measures is the conserved momentum divided by a. With $p_{\text{local}} = h/\lambda$, deduce the cosmological redshift $1 + z \equiv \lambda_{\text{obs}}/\lambda_{\text{emit}} = a_{\text{obs}}/a_{\text{emit}}$.

(a) $\mathcal{L} = T - U$ with $T = \frac{1}{2}m\dot{r}^2$:

$$\boxed{\mathcal{L}(r,\dot{r},t) = \tfrac{1}{2}m\dot{r}^2 - \frac{2\pi G}{3}\, m\rho(t)\, r^2}$$

Euler-Lagrange time! $\partial\mathcal{L}/\partial\dot{r} = m\dot{r}$ and $\partial\mathcal{L}/\partial r = -\frac{4\pi G}{3}m\rho\, r$, so $\frac{d}{dt}\partial\mathcal{L}/\partial\dot{r} = \partial\mathcal{L}/\partial r$ reads

$$\boxed{m\ddot{r} = -\frac{4\pi G}{3}\, m\rho\, r}$$

which is $m\ddot{r} = -GmM(<r)/r^2$ with $M(<r) = \frac{4\pi}{3}\rho r^3$: Newton's second law with the shell-theorem force. Neither test applies: r appears in $\mathcal{L}$ (through U), so it is not ignorable, and t appears explicitly (through $\rho(t)$), so Section 7.8's guarantee of a constant $\mathcal{H}$ is off. (A conserved quantity does exist, as (d) shows; these coordinates just do not reveal it.)

(b) With $r = ax$, the product rule gives $\dot{r} = \dot{a}x + a\dot{x}$. Substituting into (a),

$$\mathcal{L} = \tfrac{1}{2}m(\dot{a}x + a\dot{x})^2 - \frac{2\pi G}{3}\,m\rho\,a^2x^2 = \tfrac{1}{2}ma^2\dot{x}^2 + ma\dot{a}\,x\dot{x} + \tfrac{1}{2}m\dot{a}^2x^2 - \frac{2\pi G}{3}\,m\rho\,a^2x^2.$$

The cross term: by the product rule (and the chain rule for $\frac{d}{dt}x^2 = 2x\dot{x}$),

$$\frac{d}{dt}\Big(\tfrac{1}{2}ma\dot{a}\,x^2\Big) = \tfrac{1}{2}m(\dot{a}^2 + a\ddot{a})\,x^2 + ma\dot{a}\,x\dot{x} \quad\Longrightarrow\quad ma\dot{a}\,x\dot{x} = \frac{d}{dt}\Big(\tfrac{1}{2}ma\dot{a}\,x^2\Big) - \tfrac{1}{2}m\dot{a}^2x^2 - \tfrac{1}{2}ma\ddot{a}\,x^2.$$

Put that in:

$$\mathcal{L} = \tfrac{1}{2}ma^2\dot{x}^2 + \frac{d}{dt}\Big(\tfrac{1}{2}ma\dot{a}\,x^2\Big) - \tfrac{1}{2}m\dot{a}^2x^2 - \tfrac{1}{2}ma\ddot{a}\,x^2 + \tfrac{1}{2}m\dot{a}^2x^2 - \frac{2\pi G}{3}\,m\rho\,a^2x^2.$$

The two $\frac{1}{2}m\dot{a}^2x^2$ terms cancel. Friedmann, $\ddot{a} = -\frac{4\pi G}{3}\rho\,a$, turns the $a\ddot{a}$ term into $-\frac{1}{2}ma\big(-\frac{4\pi G}{3}\rho\,a\big)x^2 = +\frac{2\pi G}{3}\,m\rho\,a^2x^2$, which cancels the potential term exactly. What is left is

$$\boxed{\mathcal{L} = \tfrac{1}{2}ma^2\dot{x}^2 + \frac{d}{dt}\Big(\tfrac{1}{2}ma\dot{a}\,x^2\Big)}$$

(The whole of U went into the $a\ddot{a}$ term: the expansion is *caused* by that gravity, so in coordinates that expand with it, the gravity is already accounted for.)

(c) Write $F = \frac{1}{2}ma\dot{a}\,x^2$, a function of x and t only. Adding dF/dt to $\mathcal{L}$ adds $\int_{t_1}^{t_2}(dF/dt)\,dt = F(t_2) - F(t_1)$ to the action, and Hamilton's principle holds $x(t_1)$ and $x(t_2)$ fixed, so that number is the same for every path being compared. The stationary path, and so the equation of motion, is unchanged. (Problem 7.48 checks the same thing at the level of Lagrange's equations: $\partial(dF/dt)/\partial x$ and $\frac{d}{dt}\partial(dF/dt)/\partial\dot{x}$ are both $\partial_x\partial_t F + \dot{x}\,\partial_x^2 F$, so they cancel.)

(d) With $\mathcal{L} = \frac{1}{2}ma^2\dot{x}^2$, x does not appear: x is ignorable, and

$$\boxed{p \equiv \frac{\partial\mathcal{L}}{\partial\dot{x}} = ma^2\dot{x} \text{ is conserved}}$$

since Lagrange's equation reads $\frac{d}{dt}(ma^2\dot{x}) = \partial\mathcal{L}/\partial x = 0$. In terms of the peculiar velocity, $p = ma\,(a\dot{x}) = ma\,v_{\text{pec}}$, and p is constant, so

$$\boxed{v_{\text{pec}} \propto \frac{1}{a}}$$

Double the universe, halve every peculiar velocity.

(e) With $p = ma^2\dot{x}$,

$$\mathcal{H} = p\dot{x} - \mathcal{L} = ma^2\dot{x}^2 - \tfrac{1}{2}ma^2\dot{x}^2 = \tfrac{1}{2}ma^2\dot{x}^2 = \boxed{\tfrac{1}{2}mv_{\text{pec}}^2}$$

the kinetic energy in the peculiar velocity. It is not $T + U$: $T = \frac{1}{2}m\dot{r}^2 = \frac{1}{2}m(\dot{a}x + a\dot{x})^2$ contains the Hubble-flow part, and U is not in $\mathcal{H}$ at all. That is allowed, because the relation $r = a(t)x$ between the generalized coordinate and the position involves t (Section 7.8's condition for $\mathcal{H} = T + U$ fails). It is not conserved either: $\mathcal{L}$ depends on t through a, so by the chain rule $\partial\mathcal{L}/\partial t = \frac{1}{2}m(2a\dot{a})\dot{x}^2 = ma\dot{a}\,\dot{x}^2$, and

$$\frac{d\mathcal{H}}{dt} = -\frac{\partial\mathcal{L}}{\partial t} = -ma\dot{a}\,\dot{x}^2 = -2\,\frac{\dot{a}}{a}\cdot\tfrac{1}{2}ma^2\dot{x}^2 = -2\,\frac{\dot{a}}{a}\,\mathcal{H} \quad\Longrightarrow\quad \boxed{\mathcal{H} \propto \frac{1}{a^2}}$$

($d\ln\mathcal{H}/dt = -2\,d\ln a/dt$ integrates to $\ln\mathcal{H} = -2\ln a + \text{const.}$) That agrees with (d): $\mathcal{H} = \frac{1}{2}mv_{\text{pec}}^2$ and $v_{\text{pec}} \propto 1/a$.

(f) The local momentum is p/a with p constant, so $p_{\text{local}} \propto 1/a$. For a photon $p_{\text{local}} = h/\lambda$, so $h/\lambda \propto 1/a$, i.e. $\lambda \propto a$, and

$$\boxed{1 + z = \frac{\lambda_{\text{obs}}}{\lambda_{\text{emit}}} = \frac{a_{\text{obs}}}{a_{\text{emit}}}}$$

A photon emitted when the universe was half its present size arrives with twice its wavelength: $z = 1$.

Assess: The two results are the same statement about the same symmetry. In comoving coordinates nothing depends on x, so the momentum conjugate to x is conserved; measured locally that momentum falls as $1/a$, which is Hubble drag for a galaxy and the redshift for a photon. Energy is not conserved, and the formalism says exactly why: t is in $\mathcal{L}$, the expanding universe has no time-translation symmetry, and $d\mathcal{H}/dt = -\partial\mathcal{L}/\partial t$ gives the rate. The one hand-wave is the shell theorem on an infinite medium; general relativity makes that honest and gives (f) without borrowing.